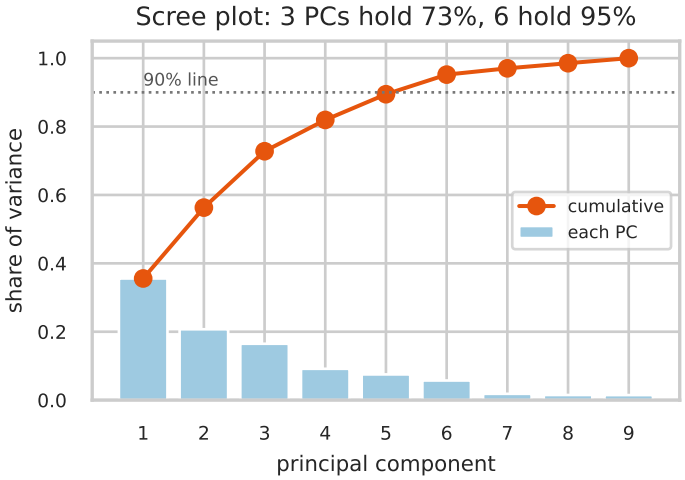
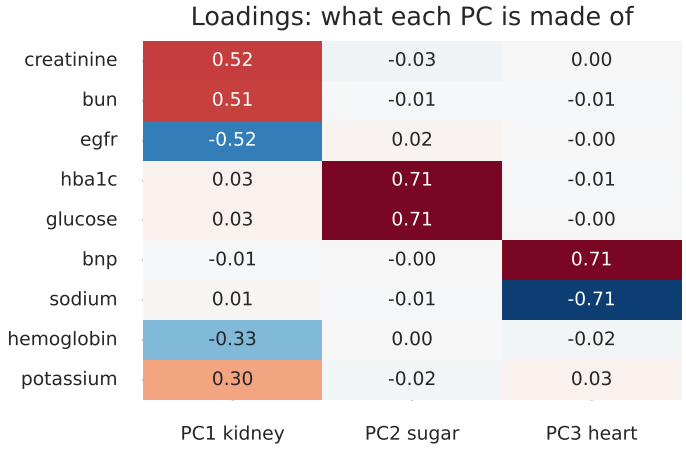
Day 20 Cheat Sheet: PCA and Dimensionality Reduction

Week 3: correlated lab panels, scree plots, loadings, and PCA inside a pipeline

TERM	WHAT IT MEANS	CODE
Dimensionality	number of columns; many related columns = redundancy	df.shape[1]
PCA	new axes (PCs) = weighted mixes of the scaled columns	PCA(n_components=k)
Principal component	direction of most remaining spread; PCs are uncorrelated	pca.components_
Explained variance	share of total spread each PC captures	explained_variance_ratio_
Scree plot	bar per PC + cumulative line; look for the drop-off	plot evr, evr.cumsum()
Loadings	weight of each original column in a PC; name the PC	components_.T
Scores	each patient's position on the new axes	pca.transform(Z)
Scale first	PCA chases variance, so units would decide the axes	StandardScaler()
Pipeline	fit scaler + PCA on train folds only (no leakage)	make_pipeline(...)
Unsupervised caveat	biggest variance is not the same as most predictive	compare CV AUC





THIS DATA: the same 4,000 simulated discharges as days 14-19, plus a simulated 9-test lab panel (seed 20)
Correlated families: creatinine, BUN and eGFR (r about 0.85); HbA1c and glucose (r 0.87); BNP and sodium (r -0.48)
Lab PCA: PC1 36% (kidney), PC2 21% (blood sugar), PC3 16% (heart); 3 PCs = 73%, 6 PCs reach 90%
Mean PC2 score: -0.46 without diabetes vs 0.94 with diabetes
Prediction (15 columns, 5-fold CV AUC): 3 PCs 0.580, 5 PCs 0.599, 8 PCs 0.648, no PCA 0.641

HABITS THAT KEEP YOU HONEST

- Scale first, then PCA; otherwise BNP (sd ~100) alone becomes PC1 and creatinine (sd ~0.3) is ignored
- Name components from loadings, and say the sign is arbitrary (a PC can flip between runs)
- PCA ignores the outcome: the most varied directions may not predict readmission at all
- Put StandardScaler and PCA inside a Pipeline so cross-validation never peeks at the test fold
- Use PCA for plots, compressing redundant panels, or speed; not by default for tree models

```
Z = StandardScaler().fit_transform(df[lab_cols])
pca = PCA().fit(Z); print(pca.explained_variance_ratio_.cumsum())
pd.DataFrame(pca.components_[:3].T, index=lab_cols)
make_pipeline(StandardScaler(), PCA(5), LogisticRegression())
```

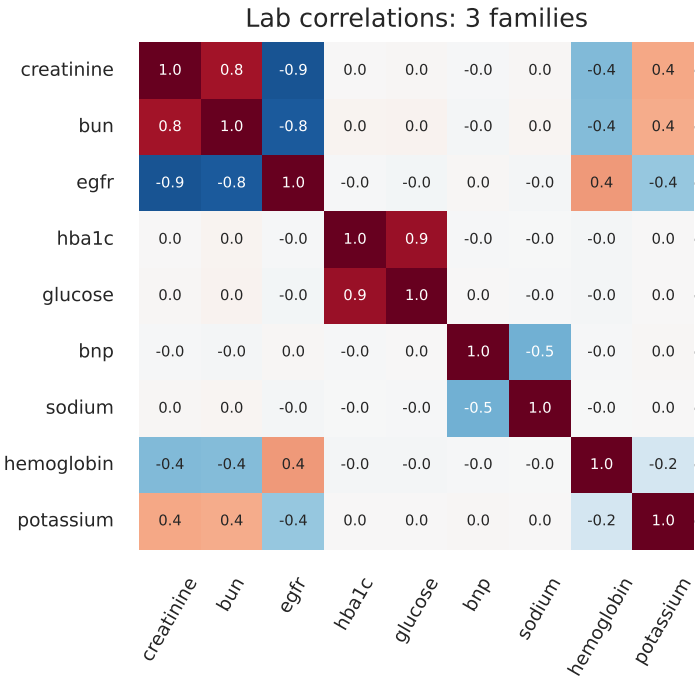
Infographic: Squashing 9 Lab Tests into 3 Meaningful Axes

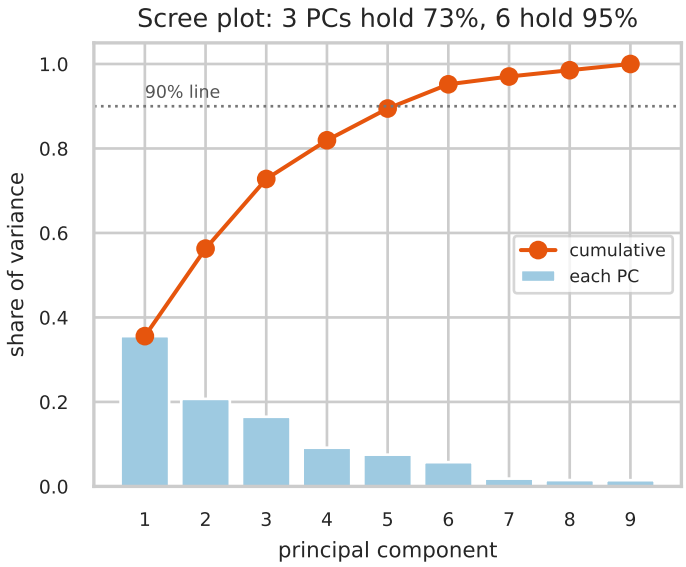
Same simulated patients, plus a lab panel. PCA finds the directions where patients differ most

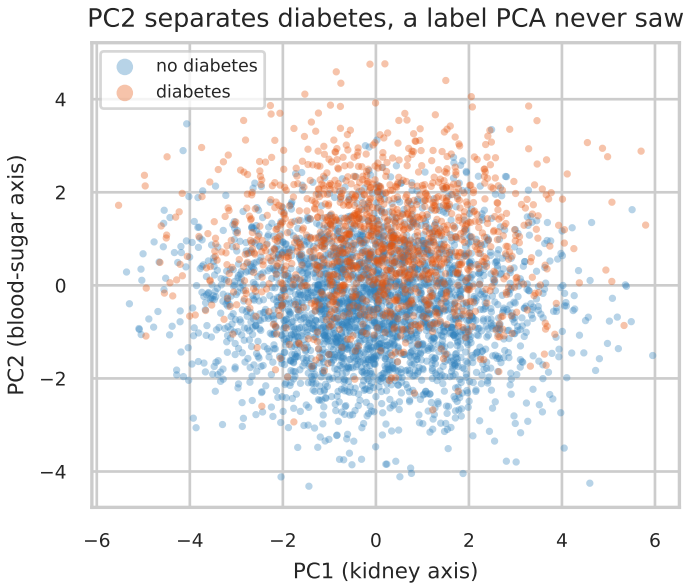
PC1 KIDNEY
36% of variance

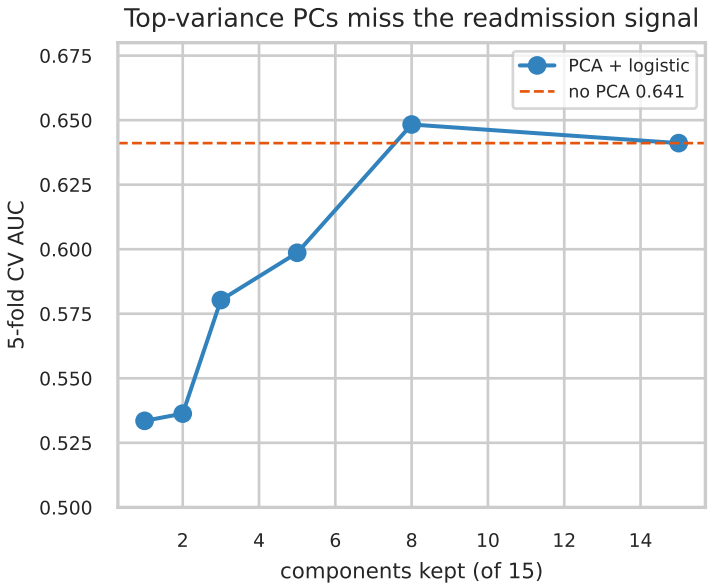
PC2 BLOOD SUGAR
21% of variance

PC3 HEART
16% of variance









Takeaways
- Related labs move together, so 9 columns hold about 3 ideas (kidney, sugar, heart)
- Loadings let you name each axis in clinical words; scores give every patient a position
- PC2 separates diabetes even though PCA never saw that column: real structure, found without labels
- But for readmission, 3 PCs of all 15 columns reach AUC 0.58 vs 0.64 with no PCA: variance is not signal
- Tomorrow: week 3 wrap-up, comparing every model so far in one honest scorecard

Hands-On Worksheet: ds-genai-90-days / day20

60-90 minutes | notebook: day20/pca_labs.ipynb + day20/README.md

[ ] mkdir day20, conda activate ds90, open day20/pca_labs.ipynb

[ ] Rebuild the days 14-19 data (snippet 1)

[ ] Add the simulated 9-test lab panel (snippet 2)

[ ] Plot the lab correlation heatmap; circle the 3 families

[ ] Scale the labs, fit PCA, print variance per PC (snippet 3)

[ ] Draw the scree plot; how many PCs reach 90%?

[ ] Print loadings for PC1-PC3 and name each one (snippet 4)

[ ] Scatter PC1 vs PC2 colored by diabetes

[ ] Try PCA WITHOUT scaling: which lab takes over PC1?

[ ] Pipeline: scaler + PCA + logistic, CV AUC for k = 1..15 (snippet 5)

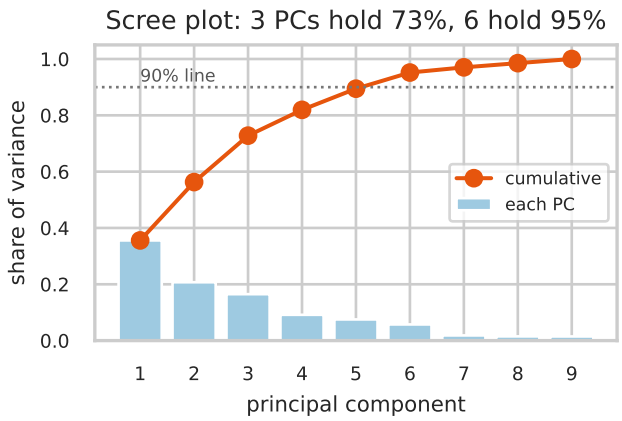
[ ] Compare with no PCA; write 2 sentences on why top PCs miss signal

[ ] README: 3 charts, a loadings table, and when you WOULD use PCA

[ ] git add day20 && git commit && git push (snippet 6)

[ ] Stretch: lab panel summary in Snowflake SQL (snippet 7)

```
Z = StandardScaler().fit_transform(
        df[lab_cols])
pca = PCA().fit(Z)
print(pca.explained_variance_ratio_
         .cumsum().round(3))
```



PC1 share: _____ PC2: _____ PC3: _____ PCs for 90%: ____ Unscaled PC1 dominated by: _____

CV AUC with 3 PCs: _____ 5 PCs: _____ 8 PCs: _____ No PCA: _____

Day 20 Self-Check Quiz

1. In one sentence, what does PCA create from the original columns?

_____

2. Why must you scale the lab values before running PCA?

_____

3. What does a scree plot show, and how do you pick the number of PCs?

_____

4. PC1 has big weights on creatinine and BUN, and a big negative one on eGFR. Name it.

_____

5. Why can the sign of a component flip between runs, and why is that fine?

_____

6. Why put StandardScaler and PCA inside a Pipeline before cross-validation?

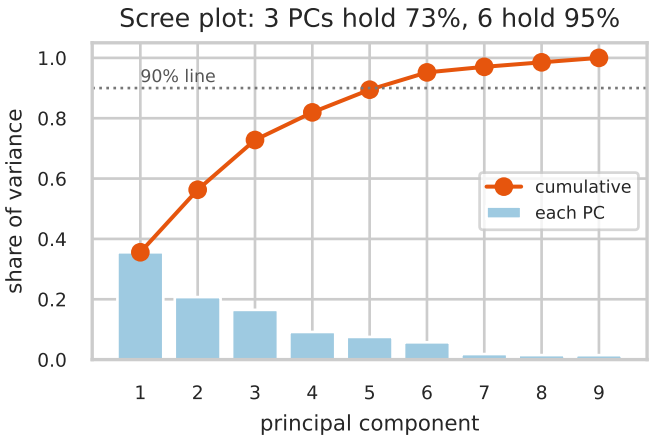
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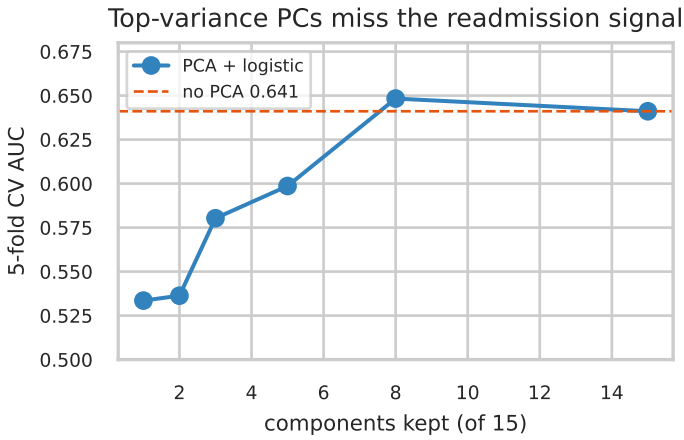
7. Keeping only 3 PCs predicts readmission worse than using all 15 columns. Why?

_____

8. Name two good uses of PCA in a healthcare analytics job.

_____





Answers:
1 new uncorrelated axes, each a weighted mix of the scaled columns, ordered by how much spread they capture
2 PCA chases variance in raw units, so BNP (sd ~100) alone becomes PC1 unscaled, just because of its units
3 the share of variance per PC and the running total; keep PCs up to the drop-off or a target such as 90% (6 here)
4 a kidney function axis: high creatinine and BUN with low eGFR means worse kidney function
5 a direction and its opposite describe the same line; read the relative weights, not the sign
6 so the scaler and PCA are fit only on training folds; fitting on all data leaks test information
7 PCA ignores the outcome; the directions with the most variance (mostly labs) are not where readmission risk lives (prior admits, follow-up)
8 compress a redundant panel into a few scores; 2-D plots of patients; speed up or de-correlate inputs for linear models